Minimum engineering requirements for assessing mining projects

D. Larsen, R. Addison, R. Kehmeier, T. Swendseid, T. Brown and J. Uhrie
RungePincockMinarco, Greenwood Village, CO, USA

Abstract

The development of a mining project necessarily progresses through many steps, beginning with the identification of a mineral resource and continuing through project construction. Several classifications of engineering study development are described, including preliminary economic assessment (PEA) or scoping studies, prefeasibility studies (PFS), feasibility studies (FS) and regulatory codes, each with a specific project organizational objective. This paper addresses the PEA, PFS and FS types of reports that are used by companies to evaluate projects, meet regulatory requirements and seek financing. The minimum requirements for engineering studies, focusing on the needs of financial institutions considering investment in a project, are outlined.

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Introduction

The financing of mining projects in recent years has come under increased scrutiny by financial institutions owing primarily to an increasing percentage of projects whose capital expenditures had overrun project budgets or had failed to produce the economic returns anticipated during the project development process. Often, unfavorable project outcomes are not the result of widespread failures of the project engineering process but arise from inadequate engineering in limited specific areas of the project scope, such as overestimated mineral reserve, inadequate mine design, insufficient metallurgical testing or underestimated capital costs. Mill facilities and process infrastructure typically are responsible for up to 80 percent of the capital expenditures required by greenfield mining projects. They are often the project areas where capital costs are inadequately developed, and consequently are the areas highly scrutinized by financial institutions. This paper offers a perspective on the level of engineering expected by financial institutions as a prerequisite to committing support to developing mining projects.

The evaluation of a mining project from exploration through development and production is a lengthy and complicated process. Mine development commitment activities for a potential project are initiated when a mineral resource is identified and continue through to the start of construction. The technical feasibility and the economic viability of each project are determined during the phases of mine development with more detailed engineering data required at each stage. There are at least four levels of engineering studies during development that are commonly acknowledged by the mining industry:

1. Conceptual study.
2. Preliminary economic assessment (PEA), or scoping study.
3. Prefeasibility study (PFS).
4. Feasibility study (FS).

Some other commonly used study names include definitive feasibility study (DFS), bankable feasibility study (BFS), front end engineering design (FEED), stage gates studies, front end loading (FEL) studies, basic engineering and detailed engineering.

In this paper, we are addressing only the PEA, PFS and FS types of reports that are used by companies to evaluate projects, meet regulatory requirements and seek financing. These are key studies to the industry, as reporting jurisdictions use them to define the point at which "ore reserves" can be publically reported. Some of the other types of reports and stages have different meanings for different companies, and many of the definitions are based on factors such as the company's incorporation type, public versus nonpublic; its size, major versus junior; where the mining project is based and what its historical reporting requirements entailed, for example, in the former Soviet Union versus China versus South America.

This paper and the Appendix Table list RungePincock-Minarco's (RPM) guidelines for the minimum suggested

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requirements for the three key levels of engineering studies, regardless of jurisdiction, incorporation or geographic location. We highlight the word "minimum" and recognize that many studies exceed our suggested guidelines. We also recognize that other companies and other engineering firms suggest different minimums from the ones listed here.

Standards or codes have been developed in many jurisdictions around the world to ensure factual and consistent reporting of mineral data is practiced. Most or all of these codes have adopted similar definitions for data collection, resource and reserve reporting, and the other aspects of evaluating mineral deposits in advancing stages of evaluation. Two common codes used in the industry are (1) the Canadian CIM Definition Standards for Mineral Resources and Mineral Reserves, or CIM reporting code, with the associated Canadian National Instrument 43-101, or NI 43-101, and (2) the Australasian Code for Reporting of Exploration Results, Mineral Resources and Ore Reserves, or JORC code. Neither of these two codes should be confused with an engineering study as they have been developed to ensure that misleading, erroneous or fraudulent information relating to mineral properties is not reported to investors on the stock exchanges they oversee.

Included in most standards are references to definitions of PEA or scoping study, PFS and FS, and the following descriptions are consistent with those definitions. As the level of engineering detail increases through each study stage, as measured by accuracy, the overall degree of uncertainty decreases.

Preliminary economic assessment/scoping study

The PEA, also commonly referred to as a scoping study, is an early-level study and the preliminary evaluation of the mining project. The principal parameters for a conceptual study are mostly assumed and/or factored. Accordingly, the level of accuracy of capital costs is as low as ±50 percent. However, the level of drilling and sampling with the required quality assurance and quality control (QA/QC) must be sufficient to define as a minimum an inferred resource. Flowsheet development, cost estimation and production scheduling are often based on limited data, test work and engineering design. Metallurgical test work is based on ore types and examines mineralogical, comminution and separation characteristics in order to establish the likely flowsheet and product recoveries. The results of a PEA typically identify:

- Probable mining and processing schemes.
- Technical parameters requiring additional examination or test work.
- General features and parameters of the proposed project.
- Magnitude of capital and operating cost estimates.
- Level of effort for project development.

A PEA is useful as a tool to determine if subsequent exploration activities and engineering studies are warranted. However, it is not valid for economic decision-making nor is it sufficient for reserve reporting.

Prefeasibility study

The PFS is an intermediate step in the engineering process to evaluate the technical and economic viability of a mining project. The prefeasibility study is a critical step for project development as it represents the minimum prerequisite for conversion of a geologic resource into a reportable reserve.

Total engineering at the prefeasibility level is still limited, often representing less than 10 percent of the total engineering effort, but should increase the level of accuracy in the cost estimate to ± 25 percent. The engineering objectives of a prefeasibility study are to study a range of development options to reasonably assess:

- Mining method and production rates.
- Dilution and extraction estimates.
- Processing method, processing rates and recovery estimate.
- Tailing and waste containment.
- Hydrology studies.
- Marketing requirements.
- Environmental and permitting requirements.
- Social license.
- Governmental requirements.
- Legal concerns.
- Detailed financial analysis and project economics with sensitivities.
- Capital cost estimates.
- Operating cost estimates.

At the prefeasibility study stage, adequate geology, drilling, sampling, QA/QC to verify the accuracy and precision of the assay test work, and mine engineering work will have been conducted to define a resource that may be convertible to a mineral reserve pending positive economic outcome of the prefeasibility study. Significant metallurgical test work, typically involving locked cycle testing (LCT) and possibly including pilot testing, will have been performed on representative ore blends of all significant rock and mineralization types based on selected composites, representative of key years in the life-of-mine schedule. In addition, the feed grade-recovery relationship for major ore types and thus blends is established. Mining and processing parameters must be sufficient for flowsheet development, production and development scheduling and major equipment selection. Typically, flowsheet modeling is conducted, particularly for the comminution circuit. With regard to infrastructure and planned operations, potential issues with important environmental and social considerations must be assessed to determine potential showstoppers, such as the existence of critical habitat and the level of involuntary resettlement required for the project. Capital and operating cost estimates use significant vendor quotes on major equipment, but other construction costs are often factored. The economic analysis of a prefeasibility study is of sufficient accuracy to assess various development options and overall project viability. However, cost estimates and engineering parameters are not typically considered of sufficient accuracy for project execution decisions or debt financing commitments.

The Canadian Institute of Mining, Metallurgy and Petroleum (CIM) defines a prefeasibility study as "a comprehensive study of a range of options for the technical and economic viability of a mineral project that has advanced to a stage where a preferred mining method, in the case of underground mining, or processing is determined. It includes a financial analysis based on reasonable assumptions on the Modifying Factors and the evaluation of any other relevant factors which are sufficient for a Qualified Person, acting reasonably, to determine if all or part of the Mineral Resource may be converted to a Mineral Reserve at the time of reporting. A Pre-feasibility Study is at a lower confidence level than a Feasibility Study." (CIM, 2014)

It is intended that through trade-off studies at the pre-

feasibility study stage, the favored development options are selected prior to commencing the feasibility study.

Feasibility study

A FS represents the next and, thus far, most detailed step in the engineering process for evaluating a mining project for a "go/no-go" decision and financing purposes. Principal parameters for a feasibility study are based on sound and complete engineering and test work. Cost estimate accuracy is higher than the prefeasibility study and is typically ±15 percent. Feasibility study objectives are the same as those previously listed for the prefeasibility study, but the levels of detail and accuracy for each objective are more stringent. In the past, the term "bankable" was used in describing a feasibility study. This term simply implied that the level of detail of the study is sufficient for potential project financing.

Detailed geologic and mine engineering work will have been conducted to define a resource and reserve. Detailed test work will have been completed to develop all mining and processing parameters for pit slope design; hydrology; geotechnical; flowsheet development; equipment selection and sizing; consumables and power consumption; material balance; general arrangement drawings; production and development schedules; and capital and operating cost estimates. Capital and operating cost estimates are derived from take-offs and full vendor quotes. A draft environmental social impact assessment (EISA) will have been submitted to regulatory authorities or is close to being submitted. The right to mine will have been granted, which is required by the U.S. Securities and Exchange Commission (SEC) to declare reserves, or can be reasonably expected to be granted, which is allowed by NI 43-101 and JORC to declare reserves. Economic analysis with sensitivities is based on annual cash flow calculations for the mine life.

The CIM defines a feasibility study as "a comprehensive technical and economic study of the selected development option for a mineral project that includes appropriately detailed assessments of applicable Modifying Factors together with any other relevant operational factors and detailed financial analysis that are necessary to demonstrate, at the time of reporting, that extraction is reasonably justified (economically mineable). The results of the study may reasonably serve as the basis for a final decision by a proponent or financial institution to proceed with, or finance, the development of the project. The confidence level of the study will be higher than that of a Pre-Feasibility Study." (CIM, 2014)

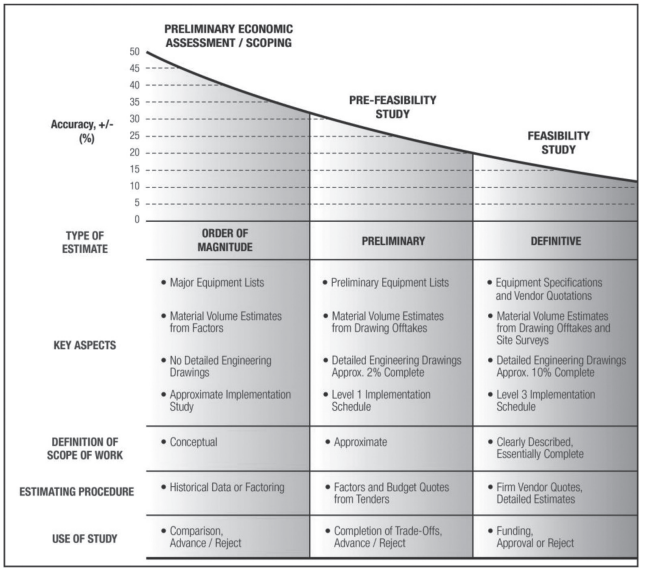


Figure 1 — Summary comparison of capital cost estimates by study level.

Key engineering study tasks

The following sections contain a brief description of the major areas that are incorporated into engineering studies. As the level of the study increases in complexity, the level of detail required increases as shown in the appended table. For a summary comparison of capital cost estimates by study level, see Fig.1.

Geology, resources and reserves. Every mineral deposit has its own unique geologic characteristics that must be considered in the preparation of engineering studies. The amount of required geologic information for the determination of resources varies significantly as a function of the engineering study level and the complexity of the deposit and mineralogy. Geologic features control economic mineralization, and with the appropriate geologic modeling, a reliable grade estimate will be determined using a combination of geologic controls and geostatistics.

As part of the geological study, which generally includes drilling, sampling and assaying, a rigorous and well-documented QA/QC program must be conducted to allow verification of the accuracy and precision of the basic sampling and assay data. Industry standards generally require a series of blanks, standard reference material, coarse and pulp duplicates, and samples to be sent to a second laboratory for assaying.

Resource estimation is based on the development of a three-dimensional model of the deposit geology, mineralization characteristics and continuity. The completed resource allows for rapid tabulation of mineral inventory and provides a basis for all subsequent determinations of reserves, mine design and planning. The objective is to provide the most reliable and accurate resource estimate with available data. The resource and reserve estimate is classified according to internationally recognized standards. The current listing rules for mining projects, minimum project stage, resource/reserve reporting code and level of engineering study required by various jurisdictions to declare "reserves" is summarized in Table 1.

Mining. Upon completion of the deposit's geologic interpretation and resource estimate, the mining method, either surface and/or underground, is selected. This selection is based on the geometry of the deposit and depth of the deposit. As the studies progress, the detail incorporated into the actual design of the mine increases.

Optimization software is then used to determine openpit economic limits or define stope shapes for underground mining projects. The selection of economic mining limits is often supported by a strategic analysis that includes assessing the impact of changes in prices, costs and recoveries. At this stage, including inferred resources in the pit limit optimization analysis can also help a company determine where to focus exploration drilling to increase measured and indicated resources. Although the strategic analysis task involves preparing preliminary production schedules, the basis lacks key design features, such as haulage ramps in the case of open pits, and therefore the results cannot be equated to declarable mineral reserves.

As the level of engineering study is advanced, the detail

Table 1 — Summary of rules for mining projects by various jurisdictions to declare reserves.

Listing entity	Listing rules that apply to mining projects	Minimum project status for listing	Minerals resource/ reserve reporting code	Study level for reserve declaration
Australian Stock Exchange.	Chapter 5.	Exploration.	JORC.	PFS.
Hong Kong Exchange.	Chapter 8.	Predevelop-ment.	JORC or CIM or SAMREC.	PFS.
Singapore Exchange.	Main Board Listing Rules 624,749, 750, 1014(2) and 1207(21; Practice Note 6.3 Appendix 7.5.	Exploration.	JORC.	PFS.
Catalist.	Catalist Listing Rules 440, 441, 704(35), 705(7), 1014(2) and 1204(23). Practice Note 4C Appendix 4C.	Exploration.	JORC.	PFS.
Shanghai Stock Exchange.	None.	None.	Classification for resources/reserves of solid mineral commodities (GB/T 17766-1999).	Chinese Standards.
TSX Venture Exchange.	NI 43-101/Appendix 3F.	Exploration.	CIM.	PFS.
New York Stock Exchange.		Advanced development.	SEC Industry Guide 7.	FS.
AIM - London Stock Exchange.	AIM Rules Guidance Note 2009.	Exploration.	JORC or CIM or SAMREC.	PFS.
London Stock Exchange.	Financial Conduct Authority Handbook (includes Main Board and AIM Listing, Prospectus and Disclosure Rules); ESMA Prospectus Directive 2011; ESMA Note on Prospectus Directive.	Predevelop-ment.	JORC or CIM or SAMREC.	PFS.

incorporated into the mine design, production scheduling, and capital and operating cost estimation increases. For the mine design, the detail and quantity of geotechnical data is critical as it dictates the pit slopes in surface mines and the design of underground openings and specific underground mining methods. Mine design and scheduling can also be affected by the need to meet processing requirements such as maintaining mill feed grades or rock type blends, and by environmental requirements such as surface and groundwater management. As knowledge of the project is increased, the mine design and production schedule can become more refined and detailed. With increasing engineering detail, resources may then be converted to reserves.

Process engineering. Process development requirements at the feasibility level should include test work on representative samples sufficient to develop a flowsheet, pilot testing of this flowsheet on representative blends of all significant ore types, and variability testing of all significant ore types and mining blends sufficient to identify probable plant throughput and recovery variation. Processing facilities are designed to produce marketable products for shipment directly to the consumers, for example, copper cathodes from solvent extraction-electrowinning (SX-EW) or to subsequent processing facilities, for eample, concentrates to smelters-refineries.

The key components of process engineering in engineering studies include:

- Metallurgical test work.
- Mineralogical studies.
- Consideration of project site conditions.
- Identification of best tailing containment location and form.
- Selection of processing flowsheet and design basis.
- Determination of processing design criteria and description.
- Plant processing facilities layout.
- Equipment sizes and specifications.
- Plant services.

Infrastructure. The infrastructure requirements for mining projects are site specific. The capital cost for infrastructure can vary substantially from site to site as a percentage of the total capital cost, and are often more of a function of the location rather than the mining or processing methods. Thus, the capital cost estimate in engineering studies must be based on a proper identification and assessment of the infrastructure requirements. Infrastructure covers a wide range of facilities and services as listed below:

- Access and service roads.
- Utilities.
- Water supply.
- Communications.
- Port and marine.

- Fuels.
- Waste disposal systems.
- Administration facilities.
- Industrial facilities.
- Transportation.
- Townsite/camp facilities.

Marketing study. Marketing studies are critical to define the nature of the market. The factors include the preferred production rate, the potential for substitution, potential competitors in the form of new mines coming onstream, product specifications, future product prices, likely buyers and terms of sales. Depending upon the commodity type, product samples are supplied to potential buyers for evaluation as part of the negotiation process. This may entail additional pilot plant test work in order to produce sufficient samples.

Environmental and social management. Environmental scientists and regulatory specialists help develop effective and economical environmental controls for mining operations, which comply with applicable environmental regulations, including international, federal, state and local, affecting the mining industry. Environmental and social considerations faced by the global mining industry in the development of mining prospects include the following general categories:

Environmental management.
- Design and implementation of baseline studies for the environmental aspects of the project.
- Development of ESIAs that address all potential impacts and propose corresponding mitigations.
- Mine permitting at the local, state, federal and international levels.
- Air quality and noise assessment.
- Mine-waste management design and associated mitigation measures required to protect the environment.
- Water management: surface and groundwater.
- Biodiversity management.
- Habitat assessment including critical and natural habitats.
- Wetlands mitigation/construction design, where applicable.
- Acid-rock drainage and leachable elements assessment and associated mitigation measures.
- Closure planning for the environmental aspects of the projects.

Social management.
- Design and implementation of baseline studies for the social aspects of the project.
- Development of ESIAs that address all potential impacts and propose corresponding mitigations.
- Stakeholder engagement actions.
- Grievance mechanism activities.
- Acquisition of a "social license."
- Closure planning.

Economic analysis. Economic analysis is performed as the final step in each engineering study to provide a measure of the project's economic viability. Economic analysis is performed using conventional pro forma cash flow analysis for the mining industry incorporating the following:

- Constant or current dollars.
- Leveraged or unleveraged financing.
- Project basis: stand alone or combined.
- Pre- or after-tax basis.
- Discounting period of project's annual cash flows: mid-year or end-of-year.

Economic measures determined in the analysis typically include (1) net present values at selected discount rates, (2) discounted cash flow return on investment, (2) internal rate of return, and (4) payback period.

Inputs to the cash flow derived from the various sections of the engineering study include (1) mining rates, production and grade schedules, (2) commodity recoveries and product schedule, (3) equipment operating schedules: number of units and operating hours, (4) manpower schedules, (5) capital costs: preproduction, sustaining and working, and (6) operating costs.

In addition to the inputs from the engineering study, other parameters to the pro-forma cash flow include (1) royalties: private and governmental, (2) commodity price or prices, and (3) the host country's tax regime: tax rates, incuding federal, state and provincial, as well as depreciation, depletion and so on.

Sensitivity analyses to the base case are performed to key project variables, which typically include (1) commodity price or prices, (2) commodity recovery or recoveries, (3) capital costs, (4) operating costs, and (5) currency exchange rates.

As can be seen above, the development of any level of study requires professionals with extensive mining experience in many different disciplines. Enhancing the capabilities of the study team will reduce the risk faced during the development of the property. For example, during the conceptual study, the study team may identify a fatal flaw that places the project at such risk that the project should not proceed at that particular point in time.

On the other hand, an experienced team can provide the knowledge base to optimize the project as much as possible or apply the best-available proven technology during the prefeasibility and feasibility study stages. Beyond the team's basic mine development experience is the consideration of the knowledge and experience of the project financing team and financial requirements that is required for bankable study preparation. Therefore, it is of critical importance to select the correct project team to ensure an optimal outcome and timely completion of the study. Following this approach will reduce both project development and cost risks. Additionally, following the traditional progression of the studies from conceptual to prefeasibility to feasibility generally saves time and money in the long run as critical issues can be identified and addressed early on rather than at a later stage when the impact can result in a costly project delay.

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References

Canadian Institute of Mining, Metallurgy and Petroleum, 2014, "Definition Standards: For Mineral Resources and Mineral Reserves," prepared by the CIM Standing Committee on Reserve Definitions, adopted by the CIM Council on May 10, 2014.

Appendix Table A1 — Minimum report contents for engineering studies.

Description	Preliminary economic assessment (PEA/scoping study)	Prefeasibility study (PFS)	Feasibility study (FS)
INTRODUCTION			
Location, topography and climate			
Site location map.	Basic map.	Preliminary map showing claims and boundaries.	Detailed map showing all claims and boundaries.
Topography map.	Basic map showing site topography.	Preliminary map showing site topographic features.	Detailed topographic map; aerial surveys verified with ground controls and surveys.
Property ownership.	Review of property lease.	Review of property lease; claims list provided; mineral rights known.	Property lease and rights secured and controlled; claims list and map provided; mineral rights secured.
Current status and history			
Historical chronology.	Basic presentation.	Full presentation.	Detailed presentation.
Past production (if any).	Basic presentation.	Full presentation.	Detailed presentation.
EXPLORATION AND GEOLOGY			
Geologic description			
Review.	Preliminary review.	Preliminary site-specific analysis.	Detailed site-specific analysis.
Data posting.	Review of available existing maps.	Detailed geologic mapping with cross-sections, lithology and mineralogy, and domains.	Deposit well-defined with three dimensional mapping, geologic maps, long sections, level plans, lithology and mineralogy, and domains.
Geologic assessment.	Preliminary.	Basic assessment and review.	Detailed assessment of structures/rock contacts, alteration, mineralization, deposit trends.
Mineralogical sampling and analysis.	Limited sampling; preliminary assessment.	Preliminary mineralogical sampling and analysis; preliminary mineralogical study.	Detailed mineralogical sampling and mapping; detailed mineralogical study.
Drilling, sampling and assaying			
Drill hole parameters.	Wide-spaced drilling, as appropriate.	Initial infills of wide-spaced drilling; preliminary grid patterns.	Close-spaced drilling on a detailed grid pattern to support calculated reserve categories.
Underground drilling.	Review of existing data.	Drilling, if accessible.	Detailed drilling, if accessible.
Samples.	Preliminary; some outcrop samples.	Geophysical and geotechnical sampling; test pits.	All sampling programs complete.
Drilling/assay data.	Preliminary check of existing drill hole data.	Check of drill holes (coordinates, elevations, angles, etc.), check assays, angled hole versus vertical hole comparison; assay flow diagram, dependable database.	Check of drill holes (coordinates, elevations, angles, etc.), check assays, angled hole versus vertical hole, etc., check assays, angled hole versus vertical hole comparison, twin hole drilling; assay flow diagram; validated database.

Appendix Table A1 — Minimum report contents for engineering studies (continued).

Description	Preliminary economic assessment (PEA/scoping study)	Prefeasibility study (PFS)	Feasibility study (FS)
QA/QC protocol and data.	Defined QA/QC protocol that verify sample and assay results. This protocol should include blanks, standard reference material, coarse and pulp duplicates, field duplicates and third party check assays. The verification samples should constitute a minimum of 10% of the sample stream.	Defined QA/QC protocol that verify sample and assay results. This protocol should include blanks, standard reference material, coarse and pulp duplicates, field duplicates and third party check assays. The verification samples should constitute a minimum of 10% of the sample stream.	Defined QA/QC protocol that verify sample and assay results. This protocol should include blanks, standard reference material, coarse and pulp duplicates, field duplicates and third party check assays. The verification samples should constitute a minimum of 10% of the sample stream.
Condemnation drilling.	None.	None.	Areas under waste dumps, tailings and plant drilled.
RESOURCES AND RESERVES (Internationally recognized standards)[a]			
Resources	Measured, indicated and inferred.	Measured, indicated and inferred.	Measured, indicated and inferred.
Geologic controls.	Assumed.	Established from geologic data and/or variograms.	Well established from geologic data.
Tonnage factors.	Preliminary assessment, if available.	Preliminary analysis and determinations.	Detailed analysis and determinations.
Statistical analysis.	Preliminary analysis and determinations.	Preliminary analysis and determinations.	Detailed analysis and determinations.
Geostatistical analysis.	Preliminary analysis and determinations.	Preliminary analysis and determinations.	Detailed analysis and determinations.
Pit optimization.	Pit limit optimization software.	Pit limit optimization software.	Pit limit optimization software.
Reserves.	Only resources estimated.	Proven and probable.	Proven and probable.
Calculation parameters.	Usually no reserves are estimated.	Known or estimated.	Detailed analysis and determinations, including dilution and losses.
Pit optimization.	Pit limit optimization software.	Pit limit optimization software.	Pit limit optimization software.
Cutoff grade (COG) equations.	Manually calculate to estimate minable inventory, or strategic mine planning software.	Manually calculate to estimate minable inventory (underground project), or strategic mine planning software (underground and openpit).	Manually calculate to estimate minable inventory (underground project), or strategic mine planning software (underground and openpit).
MINING			
Mining method.	Assumed between openpit and underground.	Specific method identified, generic equipment.	Method and mine plan finalized, including optimization of SMU and equipment types.
Openpit mine plan			
Pit slopes.	Assumed.	Preliminary 3D model based on preliminary estimates by rock type and basic geotechnical data.	3D model defined by geotechnical data from structural mapping and oriented core holes.
Pit design.	Simple LG cone outline of final pit.	Preliminary pit design from optimized analysis; preliminary haulroad incorporated, trade-off studies completed.	Detailed, optimized pit designs with phases and access for equipment operation.

Appendix Table A1 — Minimum report contents for engineering studies (continued).

Description	Preliminary economic assessment (PEA/scoping study)	Prefeasibility study (PFS)	Feasibility study (FS)
Waste dumps.	Simple outline of final dumps.	Preliminary design for total waste tonnage; incremental and final outline of dumps, trade-off studies completed.	Dump sites identified from geotechnical data; final waste tonnages determined with incremental phases, yearly and final dump outlined.
Underground mine plan			
Underground mine plan.	Assumed mining system; general outline of mine plan and development.	Preliminary mining system identified from geologic and geotechnical data; preliminary outline of mine plan and development including mine access.	Specific mining system identified from geologic and geotechnical data; detailed outline of mine plan and development including mine access.
All mining operations			
Production schedule.	Basic schedule based on assumed mine life.	Annual schedule for life of mine of ore and waste tonnages and grade, plant grade, recovery and production. Some sequence optimization, e.g., optimize grade or quality profile.	Detailed schedules with monthly time increments from ramp-up to steady state, then annual thereafter, for life of mine. Should match budget time-line schedule. Mulitple iterations to optimize mining sequence. Report ore/product quality and waste tonnages and grades plus plant recovery and production.
Capital cost estimate.	Order-of-magnitude, factored or from similar operations.	Preliminary equipment list; budget or historical price quotes; some factoring.	Detailed equipment list; firm price quotes for all major equipment items; all capital items identified.
Operating cost estimate.	Order-of-magnitude, factored or from similar operations.	Quantified estimates for labor, power and consumables; budget or historical price quotes for unit prices; some factoring.	Detailed engineering estimate by project area based on quotes and studies.
PROCESSING			
Ore sampling and test work.	Laboratory bench-scale data and ore characterization data used in combination with plant benchmarking to develop preliminary recovery and throughput.	Range of flowsheets with recovery and throughput developed from bench-scale testing and often verified with pilot scale work, especially for new processes.	Sampling of core for variability testing to identify ranges of throughput and recovery backed up with locked-cycle and/or pilot testing.
Process engineering and design			
Production rate and product(s).	First estimate of production rate and product(s).	Preliminary mining and processing rates and plant product(s).	Fixed mining and processing rates and plant product(s).
Design basis.	Preliminary using factored estimates.	General design basis; preliminary engineering drawings; trade-off studies completed.	Complete design basis; basic engineering drawings essentially complete.
Design concept.	Outline of design criteria and specifications incorporating area/regional climatic conditions.	Design criteria established for construction site incorporating known site climatic conditions.	Design specifications defined incorporating known site climatic conditions.

Appendix Table A1 — Minimum report contents for engineering studies (continued).

Description	Preliminary economic assessment (PEA/scoping study)	Prefeasibility study (PFS)	Feasibility study (FS)
Tailings containment.	Identify possible sites and capacities.	Identify several options and determine the best site or sites.	Finalize sites, location; develop geotechnical data for site and for tailings; generate general arrangement drawings and preliminary specifications.
Process description.	General.	Narrative; 1 to 2% of detail engineering complete.	Detailed; 5 to 15% of detail engineering complete.
Layout.	Approximate geographic locations and site map; no general arrangement drawings.	Optimization of facility locations on site map showing topography; simple general arrangement drawings of major equipment items.	Exact geographic locations on site map with topography; detailed general arrangement drawings; detailed layout of all facilities.
Flowsheets.	Assumed flowsheet from known processes; simple block diagram.	Establishment of probable flowsheet from preliminary test work data; major process flow diagrams; initial determinations of material and heat balances.	Detailed flowsheet based on comprehensive beneficiation test program, detailed equipment list; diagrams for all process flows; material and heat balances finalized.
Civil work.	Rough topographic maps; no soil conditions considered or quantities estimated.	Rough topographic maps; soil conditions report for foundation determinations; basic preliminary quantities.	Detailed topographic maps with soil conditions identified for foundation design, loadings and quantities.
Equipment specifications.	Major equipment items listed.	Preliminary listing of major equipment items with initial sizings and specifications.	Complete listing of major equipment items with detailed sizings and specifications.
Architectural.	None.	Sketches.	Exterior elevations only.
Piping/HVAC.	None.	Preliminary P&ID.	Major P&ID.
Electrical distribution.	None.	Basic one-line diagram.	All design one-line diagram.
Motors.	None.	General description.	Detailed list of major items with horsepower.
Instrumentation.	None.	General description.	Detailed list of components.
INFRASTRUCTURE			
Facilities.	General overview with types of support facilities described.	All required support facilities identified, sizes and quantities estimated.	All necessary support facilities identified, sized and costed.
Communications.	Communications requirements identified.	Communications systems study.	Communications licensing and standards known.
Power.	Overview of power availability and regional unit power costs.	Power sources and requirements identified; unit costs obtained from power source.	Power requirements and unit costs derived from detailed engineering study; unit costs from quotes.
HYDROLOGY			
Water sources.	Estimated using regional data.	Preliminary hydrology study.	Specific water source identified.
Water usage.	Factored plant volume and unit costs.	Required plant water volumes and unit costs estimated.	Requisite plant volumes and unit costs derived from detailed engineering/geotechnical studies.
Dewatering.	Dewatering parameters identified.	Dewatering parameters estimated.	Dewatering parameters confirmed and plan defined.

Appendix Table A1 — Minimum report contents for engineering studies (continued).

Description	Preliminary economic assessment (PEA/scoping study)	Prefeasibility study (PFS)	Feasibility study (FS)
Setting.	Preliminary evaluation of project setting for potentially significant environmental and social constraints for site data.	Preliminary evaluation of the project's impact on the environment; schedule of environmental and social other permitting requirements; evaluate project setting for potentially significant environmental and social permitting constraints from site data.	Characterization of all the project's potential impacts; finalize schedule of environmental and/or other permitting requirements; evaluate project setting for potentially significant environmental and social permitting constraints.
Data.	Collect and review all available, existing data for environmental and social studies, assessments or audits; regulatory inspections, waste handling practices, management plans, and all applicable laws and regulations; social, training or safety programs identified.	Collect and review available data from existing databases for environmental studies, assessments or audits; regulatory inspections, waste handling practices; management plans; and all applicable laws and regulations; plans; initiate baseline data gathering; social, training, and health /safety programs identified.	All requisite environmental data for project are identified; site sampling and analyses are complete; detailed review of the type, scope and schedule for producing environmental and social government reports; comprehensive gathering and evaluation of baseline environmental and social conditions; social, training, and health/safety programs confirmed.
EIS/EA.	None.	Draft EIS/EA initiated.	Draft EIS/EA submitted to regulatory authorities.
Reporting and Plans.	Conceptual plans for mitigating any identified environmental issues.	Preparation of environmental and social plans and monitoring programs; preliminary sediment and erosion control plan; conceptual closure plan; evaluation of acid rock drainage; geotechnical stability review of waste dumps and tailings dam; preliminary impact mitigation plan; preliminary spill and emergency response plan.	Environmental characteristics used in project design; environmental plans and monitoring programs are finalized; sediment and erosion control plan; finalize management plans for tailings and wasterock; management plan finalized for solid and hazardous wastes; finalize impact mitigation plan; geotechnical stability analysis of all major facilities; finalize closure plan; final analysis of acid rock drainage; finalize spill and emergency response plan.
Monitoring.	Not considered.	Outline of a site environmental monitoring plan.	Complete environmental monitoring plan.
Permit requirements.	General overview.	Comprehensive overview and listing of required permits.	Detailed evaluation of all pertinent authorizations and permitting requirements and schedule for obtaining operating license.
PROJECT DEVELOPMENT SCHEDULE			
Development plan.	Development period and mine life estimated.	Development period and overall schedule estimated; mine life determined; development schedule set.	Detailed development schedule; mine life known; development schedule finalized.

Appendix Table A1 — Minimum report contents for engineering studies (continued).

Description	Preliminary economic assessment (PEA/scoping study)	Prefeasibility study (PFS)	Feasibility study (FS)
Project master schedule.	Estimated showing start and end of construction; Gantt bar chart of major work elements.	Gantt bar chart with overall time frames; schedule outline for detailed engineering; QA/QC program outlined; preliminary construction schedule; preliminary project execution plan.	Gantt bar chart with overall time frames and project flow planning; detailed project level schedule showing project deliverables and detailed engineering; CP schedule; major milestones identified; project control system outlined; QA/QC and safety program finalized; preliminary project procedures manual; project design basis finalized.
CAPITAL COST ESTIMATE			
Basis: Civil Structural. Architectural. Piping/HVAC. Electrical. Instrumentation. Construction labor. Construction labor productivity. Material volumes/amounts. Material/equipment pricing. Infrastructure.	Order-of-magnitude based historical data or factoring.	Estimates from historical factors, percentages and vendor quotes based on material volumes.	Detailed from estimates; engineering 15 to 25% complete; multiple vendor quotes.
Contractors.	Included in unit cost or as a percentage of total cost.	Percentage of direct cost by cost area for contractor; historical for subcontractors.	Written quotes from contractor and subcontractors.
EPCM.	Percentage of estimated construction cost.	Percentage of detailed construction cost.	Calculated estimate from EPC(M) firm.
Pricing.	FOB mine site including all taxes and duties.	FOB mine site including all taxes and duties.	FOB mine site including all taxes and duties.
Owner's.	Historical estimate.	Estimate from experience factored for similar project.	Estimate prepared from detail zero based budget.
Environmental compliance.	Factored from historical experience.	Estimate from experience factored for similar project.	Estimate prepared from detail zero based budget for design engineering and specific permit requirements.
Escalation.	Typically not considered.	Based on company's current budget percentage.	Based by cost area with risk.
Working capital.	Factored from historical experience.	Estimate from experience factored for similar project.	Estimate prepared from detail zero based budget.
Accuracy.	+/– 50%	+/– 25%	+/– 15%
Contingency.	25%	15%	10%
OPERATING COST ESTIMATE			
Basis.	Order-of-magnitude estimate.	Estimates for unit rates and quantified estimates with some factoring.	Detailed from zero-based budget; minimal factoring.

Appendix Table A1 — Minimum report contents for engineering studies (continued).

Description	Preliminary economic assessment (PEA/scoping study)	Prefeasibility study (PFS)	Feasibility study (FS)
Operating quantities.	General.	Quantified by estimates with some factoring.	Detailed estimates.
Unit costs.	Historical unit costs and factoring.	Estimates for labor, power and consumables; some factoring.	Letter quotes from vendors; minimal factoring.
Accuracy.	+/– 35%	+/– 25%	+/– 15%
ECONOMIC EVALUATION			
Financial analysis.	Preliminary assessment of principal economic parameters.	Assessment of the principal economic parameters.	Full assessment of all principal economic parameters.
Marketing and commodity price(s).	Industry knowledge and consensus pricing.	Preliminary market analysis to confirm product placement, quality targets and production constraints. Consensus pricing.	Detailed marketing study and consensus pricing or price forecasts.
Royalties and taxes.	Preliminary assessment.	Preliminary analysis.	Detailed analysis with tax authority opinion.
Smelting, refining and freight.	Historical data.	Budgetary quotes.	Firm quotes.
Cash flow analysis.	Simple cash flow.	Preliminary cash flow.	Formal, detailed cash flow.
Economic criteria.	Simple IRR and NPV (pre- and after-tax.	Preliminary IRR and NPV (pre- and after-tax).	Fully defined IRR, NPV, ROI and payback period (pre- and after-tax)
Sensitivity analysis.	Basic analysis to minimal amount of project variables.	Preliminary to selected key project variables.	Numerous analysis to all key project variables.
RISK EVALUATION			
Risk assessment.	General overview.	Fatal flaw analysis.	Risk workshop. Formal Monte Carlo analysis and fatal flaw analysis.
Project.	Preliminary overview of geology, engineering and environmental.	Preliminary environmental, country, permitting, technology, and business; detailed geology and engineering.	Detailed geology, engineering, environmental, legal, permitting, country, technology, business and financial.

[a]Internationally recognized standards include:

1. Canadian National Instrument 43-101 and 43-101 CP.
2. Australasian Code for Reporting of Mineral Resources and Ore Reserves, prepared by the Joint Ore Reserve Committed (JORC).
3. U.S. Securities & Exchange Commission Industry Guide 7.
4. SME Guide for Reporting Exploration Information, Mineral Resources and Mineral Reserves, by the Society for Mining, Metallurgy & Exploration, Englewood, CO.
5. Environmental aspects should consider applicable local, national and international guidelines, including IFC and Equator Principle standards.

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